

CENG381 Stochastic Processes

Renewal-Reward and Residual Life - Answer Key 1

Q1 Solution

a) Mean cycle length:

A cycle consists of one ON period and one OFF period.

ON $\sim \text{Exp}(\mu_{\text{on}})$ with mean $1/\mu_{\text{on}} = 4$ hours.

OFF $\sim \text{Exp}(\mu_{\text{off}})$ with mean $1/\mu_{\text{off}} = 1$ hour.

$$E[X] = E[\text{ON}] + E[\text{OFF}] = 4 + 1 = 5 \text{ hours per cycle}$$

b) Long-run fraction of time ON:

Define reward R_n = length of ON period in cycle n .

Since ON $\sim \text{Exp}(\mu_{\text{on}})$: $E[R] = E[\text{ON}] = 4$ hours.

Renewal-Reward Theorem:

$$\text{Long-run fraction of time ON} = E[R] / E[X] = 4 / 5 = 0.8$$

Intuition check: the machine is ON for 4 hours out of every 5-hour cycle on average, so it should be ON 80% of the time. (OK)

c) Long-run fraction of time OFF:

Define reward R_n = length of OFF period in cycle n .

Since OFF $\sim \text{Exp}(\mu_{\text{off}})$: $E[R] = E[\text{OFF}] = 1$ hour.

$$\text{Long-run fraction of time OFF} = E[R] / E[X] = 1 / 5 = 0.2$$

Check: fraction ON + fraction OFF = $0.8 + 0.2 = 1.0$ (OK)

This result also follows directly: since every moment the machine is either ON or OFF, the two fractions must sum to 1.

Q2 Solution

a) $E[X]$ and $E[X^2]$ for $X \sim \text{Uniform}[0,4]$:

$$E[X] = (0 + 4)/2 = 2 \text{ minutes}$$

$$\begin{aligned} E[X^2] &= \text{integral from 0 to 4 of } x^2 * (1/4) dx \\ &= (1/4) * [x^3/3] \text{ from 0 to 4} \end{aligned}$$

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$$= (1/4) * (64/3)$$

$$= 64/12 = 16/3 \sim 5.333 \text{ minutes}^2$$

b) Expected residual life E[B]:

Using the stationary formula for residual life:

$$E[B] = E[X^2] / (2 * E[X])$$

$$= (16/3) / (2 * 2)$$

$$= (16/3) / 4$$

$$= 4/3 \sim 1.333 \text{ minutes}$$

Despite inter-arrival times averaging 2 minutes, you expect to wait only $4/3 \sim 1.33$ minutes until the next customer. This is because your arrival is not at the START of an interval -- you land somewhere in the middle of a running interval.

c) Expected length of the interval containing t (Inspection Paradox):

The interval containing time t has expected length:

$$E[A(t) + B(t)] = E[X^2] / E[X] \quad (\text{in stationarity})$$

$$= (16/3) / 2$$

$$= 8/3 \sim 2.667 \text{ minutes}$$

Compared to $E[X] = 2$ minutes, the interval you land in is 33% LONGER.

Inspection Paradox explanation:

When you arrive at a random time t, you are more likely to land in a LONG interval than in a short one -- because longer intervals occupy more of the time axis. A gap of length L 'covers' L units of time, so the probability of landing in it is proportional to L, not to its frequency.

This biases the sampled interval upward: $E[X^2]/E[X] \geq E[X]$ with equality only when all intervals have the same length (no variance).

Q3 Solution

a) E[X], E[X^2], and Inspection Paradox check:

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$$E[X] = 2 \cdot 0.6 + 8 \cdot 0.4 = 1.2 + 3.2 = 4.4 \text{ minutes}$$

$$E[X^2] = 2^2 \cdot 0.6 + 8^2 \cdot 0.4 = 4 \cdot 0.6 + 64 \cdot 0.4 = 2.4 + 25.6 = 28.0$$

$$E[X^2]/E[X] = 28.0 / 4.4 = 70/11 \sim 6.364 \text{ minutes}$$

Compare to $E[X] = 4.4$ minutes.

$6.364 > 4.4$, so $E[X^2]/E[X] > E[X]$ -- Inspection Paradox confirmed.

The variance of X is $\text{Var}[X] = E[X^2] - (E[X])^2 = 28 - 19.36 = 8.64 > 0$,
which is exactly why $E[X^2] > (E[X])^2$, i.e., $E[X^2]/E[X] > E[X]$.

b) Probability of landing in short vs long gap:

In stationarity, the probability of arriving in a gap of type (length L , prob p_L):

$P(\text{land in gap of length } L) \text{ proportional to } p_L \cdot L$

Unnormalised weights:

Short gap ($L=2$): $0.6 \cdot 2 = 1.2$

Long gap ($L=8$): $0.4 \cdot 8 = 3.2$

Total weight = $E[X] = 4.4$

$$P(\text{land in short gap}) = 1.2 / 4.4 = 12/44 = 3/11 \sim 0.273$$

$$P(\text{land in long gap}) = 3.2 / 4.4 = 32/44 = 8/11 \sim 0.727$$

Despite short gaps being more frequent (prob $0.6 > 0.4$), you are MORE likely to land in a LONG gap (prob $0.727 > 0.273$) because long gaps are 4 times as wide and therefore catch arrivals 4 times more often per gap.

c) $E[B]$ via total expectation and verification:

If you arrive at a Uniform point within a gap of length L , the residual time B (time until the next bus) is Uniform $[0, L]$, with mean $L/2$.

$$E[B \mid \text{short gap}] = 2/2 = 1 \text{ minute}$$

$$E[B \mid \text{long gap}] = 8/2 = 4 \text{ minutes}$$

Law of total expectation:

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$$\begin{aligned}E[B] &= E[B \mid \text{short gap}]P(\text{short}) + E[B \mid \text{long gap}]P(\text{long}) \\&= 1 * (3/11) + 4 * (8/11) \\&= 3/11 + 32/11 \\&= 35/11 \sim 3.182 \text{ minutes}\end{aligned}$$

Verify using the formula:

$$E[X^2] / (2 * E[X]) = 28 / (2 * 4.4) = 28 / 8.8 = 35/11 \sim 3.182 \text{ (OK)}$$

Even though buses are spaced 4.4 minutes apart on average, you expect to wait 3.18 minutes -- more than half the average gap (2.2 minutes) -- because your arrival is disproportionately likely to land in a long gap.